SS-S-169B
June 9, 1975
SUPERSEDING
Int. Fed. Spec. SS-S-00169A (GSA-FSS)
January 5, 1968 and
Fed. Spec. SS-S-169
April 20, 1954

FEDERAL SPECIFICATION

SEALER, JOINT, SEWER, MINERAL-FILLED, HOT-POUR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for hot pour, mineral-filled joint-sealing compound used in filling joints of bell and spigot sewer pipe.

1.2 Classification. The sealing compound shall be furnished in the following types and classes, as specified (6.2):

Type I - Asphalt base.
Type II - Coal-tar base.
Type III - Plastic base.

Class 1 - Pour-in-place.
Class 2 - Die-cast, ring-and-collar.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-D-723 - Drums, Fiber.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8030

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Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publications:

C 33 - Standard Specification for Concrete Aggregates.
C 190 - Standard Methods of Test for Tensile Strength of Hydraulic Cement Mortars.
D 4 - Bitumen, Test For.
D 5 - Penetration of Bituminous Materials, Test For.
D 36 - Softening Point of Asphalts and Tar Pitches (Ring-and-Ball Apparatus) Test For.
D 113 - Ductility of Bituminous Materials, Test For.
D 140 - Bituminous Materials, Sampling.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American Association of State Highway and Transportation Officials (AASHTO) Publications:

T 43 - Specific Gravity of Bituminous Materials.
T 109 - Settlement Ratio of Mineral Matter in Bituminous Filler.

(Copies may be obtained from the American Association of State Highway and Transportation Officials, 341 National Press Building, Washington, DC 20004.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3 REQUIREMENTS

3.1 Description. Joint-sealer compounds shall consist essentially of asphalt, coal-tar pitch or plastic soluble in CS_2 and inert mineral filler. The compounds shall be free from water, uniform in appearance and consistency, and shall not foam when heated to 350° F.

3.1.1 Composition. Proportions of component materials of the compounds, expressed as percentages of total weight, shall be within the limits shown in table I.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality entirely suitable for the purpose.

3.3 Chemical resistance. No chemical action shall occur when the joint-sealer compounds are subjected to the test specified in 4.4.2.

3.4 Physical properties. The physical properties of joint-sealing compounds shall be as shown in table II.

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TABLE I. Chemical composition

	Type I		Type II		Type III
	Class 1	Class 2	Class 1	Class 2	Class 1
	Percent	Percent	Percent	Percent	Percent
Asphalt (soluble in CS_2)	45-60	40-55	-----	-----	-----
Coal-tar pitch [1]			75-80	70-80	-----
Plastic [2] (soluble in CS_2)	-----	-----	-----	-----	35-50
Inert mineral matter (determined as ash)	40-55	45-60	20-25	20-30	50-65
Organic matter (insoluble in CS_2), maximum	5	5	-----	-----	5

[1] The coal-tar pitch shall be produced from coke-oven tar by distillation or special processing.
[2] Acyclic polymeric hydrocarbon.

TABLE II. Physical properties

	Type I		Type II		Type III
	Class 1	Class 2	Class 1	Class 2	Class 1
Specific gravity at 77° F.	1.40-1.55	1.40-1.60	1.40-1.50	1.35-1.50	1.65-1.75
Ductility at 77° F. (cm.), minimum	1.55	0.5	---------	---------	---------
Settlement ratio, maximum	1.25	1.25	1.25	1.25	1.25
Softening point, °F.	190-250	240-270	180-190	210-225	200-230
Penetration: (mm.)					
At 77° F., 100 gm., 5 sec.	5-15	4-8	0-2	4-9	0-1
At 115° F., 50 gm., 5 sec.	---------	---------	5-10	12-25	0-3

3.5 Settlement ratio. The settlement ratio shall be determined in accordance with the tests specified in table III.

3.6 Bond stress. When tested as specified in 4.3.1 joint-sealing compounds shall not fail, either in bond or cohesion, under a bond stress of 75 p.s.i.

3.7 Ductility. Ductility shall be determined in accordance with the test specified in table III.

3.8 Determination of asphalt and plastic. The content of inert mineral filler, asphalt, plastic, and organic matter in the compound shall be determined in accordance with the test specified in table III.

3.9 Penetration. The penetration of the compound shall be determined in accordance with the test specified in table III.

3.10 Softening point. The softening point of the compound shall be determined in accordance with the test specified in table III.

3.11 Specific gravity. The specific gravity of the compound shall be determined in accordance with the test specified in table III.

3.12 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of commodity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling. For purposes of sampling, a lot shall be all the compound of one type and class submitted for delivery at the same time. Sampling shall be in accordance with ASTM D 140, except that 20 pounds of a representative sample shall be furnished.

4.3 Inspection of preparation for delivery requirements. The samples selected in accordance with 4.1 shall be examined to determine compliance with section 5 of this specification, and as required by specifications referenced therein.

4.4 Tests. The following tests shall be made in accordance with the methods shown in table III.

TABLE III. Test methods

Tests	Method
Ductility 1/	ASTM D 113
Inert mineral filler, asphalt, plastic soluble in CS_2 and organic matter insoluble in CS_2 2/	ASTM D 4
Penetration	ASTM D 5
Softening point	ASTM D 36
Specific gravity	AASHTO T 43
Settlement ratio	AASHTO T 109

1/ A pull of 1 cm. per minute shall be applied in lieu of that required in ASTM D 113.

2/ For type II compound, percentage of coal tar shall be considered as the difference between 100 and the percentage of inert mineral filler, provided the manufacturer has certified the compound consists only of inert mineral filler and coal tar, as required.

4.4.1 Bond stress.

4.4.1.1 Apparatus. The testing apparatus shall consist of briquet molds, testing machine, and clips in accordance with ASTM C 190.

4.4.1.2 Molding test specimens. Three portland-cement briquets shall be made and cured for 7 days in accordance with ASTM C 190, except that the sand shall be concrete sand conforming to the requirements of ASTM C 33. The briquets shall be cut smoothly in two parts transversely at the waist, oven dried at 220° F. for 2 hours and allowed to cool at room temperature for a least 2 hours. The two parts shall be placed in a suitable mold or form and a one-inch-square piece of vitrified clay with all glaze removed shall be placed between and approximately 3/8 inch from the two parts of the briquet and so that the longitudinal axis will be approximately normal to the one-inch-square surfaces. Sealing compound, heated to a free-flowing temperature but not in excess of the pouring temperature indicated on the container, shall be poured into the mold between the cut surfaces of the briquet and the piece of vitrified clay as indicated in figure 1 and allowed to air cool at room temperature for at least 2 hours. If the manufacturer's directions for use of the compound as printed on the container require priming, the cut surfaces of the briquet and the vitrified clay shall be so primed with the recommended primer before pouring the compound. Any bituminous material which may have accumulated on the briquets or pieces of vitrified clay other than on the one-inch-square surfaces shall be removed with a knife.

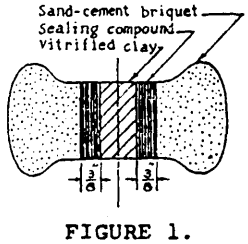


FIGURE 1.

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4.4.1.3 Test procedure. The bearing surfaces of the clips of the testing machine shall be clean and free from incrustations. The roller bearings shall be well oiled and in condition to ensure free turning. The stirrups supporting the clips shall be kept from accumulations and the pivots shall be in proper adjustment so that the clips may swing freely on the pivots without binding in the stirrups. The test specimens shall be carefully centered in the clips. The load shall be applied continuously at a uniformly increasing rate of 600 ± 25 pounds per minute until one specimen fails in bond or in cohesion.

4.4.2 Chemical resistance. The compounds shall comply with the requirements of 3.3 when immersed for 5 days in each of the following solutions: 1 percent HCL, 1 percent HNO_3, 1 percent H_2SO_4, 5 percent KOH, and saturated H_2S.

4.4.3 Tests report. The laboratory report on the tests shall state clearly the pertinent facts observed for each test, shall state whether the specimens failed in bond or in cohesion and shall state the applied load.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (6.2).

5.1.1 Level A. The class 1 sealer shall be packed in 10-gallon drums conforming with PPP-D-723, type III.

5.1.2 Level B. The class 1 sealer shall be packed in 10-gallon drums conforming with PPP-D-723, type II.

5.1.3 Level C. The class 1 sealer shall be packed to insure carrier acceptance and safe delivery at destination at lowest rates, in containers complying with the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Joint-sealer compounds are used in filling joints of bell and spigot sewer pipe. Class 1 material is the preferred material under normal-use conditions. When weather and trench conditions are such that satisfactory in-place pouring could not be effected, class 2 material should be used. Class 2 material is not recommended for general stocking purposes because of the limited conditions under which its use is recommended.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (1.2).
(c) Selection of applicable level of packing required (5.1).
(d) Whether civil or military.

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6.3 Class 1 material is the preferred material under normal-use conditions. When weather and trench conditions are such that satisfactory in place pouring could not be effected, class 2 material should be used. Class 2 material is not recommended for general stocking purposes because of the limited conditions under which its use is recommended.

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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